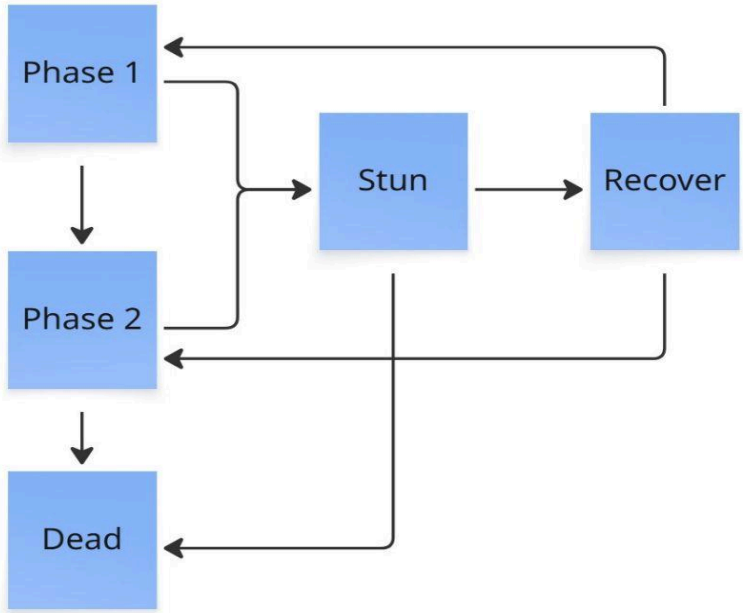
Alex DiFlaviano - A3 AI System Implementation Documentation



	Title	Behaviors	Transitions
1	Phase 1	This phase has the boss moving left and right slowly, the boss is shooting a small projectile at the player that will slow them when hit. The boss can be stunned in this state. Once its done being stunned it goes into recovery and then back into this phase. This phase has 40 health. Once that health is depleted the boss changes states.	Phase1 => Phase 2 Phase1 => Stun
2	Phase 2	This phase has the boss also moving left and right but now faster, this creates more of a challenge for the player to hit it. The boss now also shoots bigger projectiles at the player that are harder to dodge. The boss can be stunned in this state. Once its done being stunned it goes into recovery and then back into this phase. This phase also has 40 health. Once this health is depleted the boss changes to	Phase2 => Stun Phase2 => Dead
3	Stun	This phase is a smaller transition phase. All this phase does is stun the boss. It is not able to move or shoot. This phase only lasts a short amount of time before transitioning into the recovery state that handles the logic of getting back to the right Phase.	Stun => Recover Stun => Dead
4	Recover	The recover state handles knowing which Phase the boss is in and making sure that it returns back to the correct phase after being stunned.	Recover => Phase1 Recover => Phase2
5	Dead	The death phase is as it sounds, it handles when the character has no health. It is essentially the boss' game over	No transitions from Dead

Challenges faced during implementation and their solutions:

Some challenges I faced during the implementation of my AI FSM getting the agent to bounce back and forth from one point to the other. As well as my boss not switching states at all. This was pretty much due to calling the wrong state changes in too many places, as well as pretty much overwriting the state changes in the wrong places. I also had issues with the boss not shooting at all and when I was able to get that fixed it was shooting a million at once. This was due to a never ending coroutine that was fixed by adding a boolean (isShooting) that could be flipped when the boss could see the player but wasn't shooting at them. Also the stun was broken because I wasn't subtracting time from my variable properly. My collision check for the boss taking damage wasn't actually applying damage to the boss at all, this was fixed with a different function that handled the damage as well as some state switching for going from phase 1 to 2 as well as phase 2 to dead. Also stun wasn't actually locking the agent's movement. This was fixed with flagging isStopped as true. The player also couldn't shoot their projectile. Mostly due to coding the input with the old system. Once I added the button to the new input system this got working.